

Esteban Padilla Cerdio

Zurich, Switzerland — Permit B — Open to relocate

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PROFESSIONAL SUMMARY

Robotics research engineer specialized in **autonomous navigation, localization, sensor fusion, and real-time robotics**. First author of an **RSS 2026** paper on zero-shot semantic navigation, with hands-on experience in physical robot deployment. Strong **Python and C++** expertise backed by four software engineering internships at **Meta**.

PROFESSIONAL EXPERIENCE

Research Assistant

Computer Vision and Geometry Lab, ETH Zurich

Sep 2026 – Present

Zurich, Switzerland

- Implementing a **full-stack semantic SLAM pipeline** with video-based mapping, dense object localization, visual frontier navigation and **multi-robot collaboration**, under supervision of **Prof. Dr. Marc Pollefeys**.
- Setup, **deployment**, and maintenance of **Spot Arm, Reachy2, and Franka Duo** for lab experiments.

Research Intern

GVLab, The University of Tokyo

Feb 2026 – Aug 2026

Tokyo, Japan

- Developed [CommonGround](#) , a real-time **multi-agent localization and tracking** system from a grid of force plates, estimating **human and robot position**, velocity, heading, and load **without cameras or wearables**.
- Designed **probabilistic tracking, constrained optimization, and sensor-fusion** methods for footprint assignment, identity preservation and **Human-Robot Interaction analysis** in an underdetermined system.

Teaching Assistant

Soft Robotics Laboratory, ETH Zurich

Sep 2025 – Feb 2026

Zurich, Switzerland

- Implemented a **PPO training pipeline in Isaac Lab** and deployed it to an [ORCA](#)  robotic hand in support of the [Real World Robotics](#)  course under **Prof. Dr. Robert Katzschmann**.
- Collected **bimanual manipulation demos** using **motion capture** and **robot hardware** for [EgoVerse](#) .

Software Engineering Intern, PyTorch / ExecuTorch

Meta

Jun 2024 – Aug 2024

New York, NY, USA

- Built open-source **Vulkan GPU characterization** tooling; optimized **inference in edge devices** with memory-management changes, loop unrolling, and **4-bit quantization**, reducing representative inference runtimes by approximately **70%** over the previous implementation. [Repository](#) .

Software Engineering Intern, Core AI

Meta

Jun 2023 – Aug 2023

Seattle, WA, USA

- Re-engineered a **deep-learning image and video enhancement pipeline** for **C++ CPU-only execution** in servers and **AR Glasses**, reducing runtime by **99.66%** over the previous implementation.

Software Engineering Intern, WhatsApp Infrastructure

Meta

May 2022 – Jul 2022

Menlo Park, CA, USA

- Built **asynchronous crash-report processing** with logging and symbolication in **Hack**, reducing client upload wait time by approximately **95%**.

PUBLICATION

Esteban Padilla Cerdio, Boyang Sun, Marc Pollefeys, Hermann Blum. “*OpenFrontier: General Navigation with Visual-Language Grounded Frontiers*.” **Robotics: Science and Systems (RSS)**, 2026. [Publication](#)  [Website](#) 

Zero-shot object goal navigation using visual frontiers, VLM semantic grounding, and 3D object localization with **SAM3**. Achieved **77.3% success rate** in HM3D ObjectNav and **real-time execution with Spot**.

ROBOTICS PROJECTS

Dexterous Tendon-Driven Robotic Hand | Team Project, ETH Zurich

Sep 2024 – Feb 2025

- Designed and integrated a **17-DoF anthropomorphic robotic hand** with an actuated wrist on a **6-DoF arm** for high-dexterity grasping, full finger opposition, and manipulation of irregular objects.
- Implemented **motion capture, teleoperation** and **imitation-learning** workflows for autonomous object sorting with **ROS, Python**, kinematics, computer vision, and physical hardware testing. [Repository](#) .

Distributed MPC for Modular Aerial Robots | Team Project, ETH Zurich

Feb 2025 – May 2025

- Implemented an **ADMM-based distributed model predictive controller** for parallel trajectory optimization and **collision-aware coordinated motion** for the [DRAGON](#)  in simulation. [Repository](#) .

Easy Object-Conscious SLAM | Individual Project, Ulm University

Oct 2023 – Mar 2024

- Developed a modular **ROS2 semantic SLAM** system for budget-friendly platforms, combining **2D LiDAR**, monocular vision, wheel odometry, **YOLOv8**, and camera calibration on a TurtleBot3 in Webots.
- Implemented point-cloud projection, **probabilistic object tracking**, mapping and localization. [Repository](#) .

EDUCATION

ETH Zurich

Master of Science — Robotics, Systems and Control
SECIHITI Scholarship for Postgraduate Studies in Science and Humanities 2025

Sep 2024 – Aug 2026

Zurich, Switzerland

Tecnologico de Monterrey

Bachelor of Science — Robotics and Digital Systems Engineering
Valedictorian (Highest GPA, department)
Summa Cum Laude (GPA above 95/100, department)
CENEVAL National Exam Award 2024 (top 1.25%, national)

Aug 2020 – Jun 2024

Queretaro, Mexico

Ulm University

Advanced MSc Exchange — Telecommunications and Information Technology
Baden-Wuerttemberg Stipendium

Sep 2023 – Mar 2024

Ulm, Germany

SKILLS

Autonomy and Spatial Intelligence: SLAM, autonomous navigation, multi-agent localization and mapping, sensor fusion, state estimation, motion and path planning, language-grounded navigation, multi-object tracking, frontier exploration, navigation state machines

Perception: vision-language models, 3D LiDAR, RGB-D vision, point-cloud processing, object detection and segmentation, OpenCV, SAM3, YOLO, camera calibration, monocular vision, 2D-to-3D projection, force sensing

Robotics Software: ROS/ROS2, Nav2, real-time pipelines, robot bring-up, hardware testing, distributed systems, teleoperation, motion capture retargeting, Rokoko Gloves, OptiTrack Motive 3

Robot Learning: PyTorch, imitation learning, reinforcement learning, behavior cloning, sim-to-real, policy deployment, ACT, NVIDIA GR00T, LeRobot, GPU training

Simulation: Isaac Sim / Isaac Lab, MuJoCo, Gazebo, Habitat, PyBullet, Webots

Robotics Hardware: Boston Dynamics Spot Arm, Franka Panda, Pollen Robotics Reachy2, TurtleBot3, Aria glasses, Meta Quest, ORCA dexterous hand, Franka Duo, NAO Pepper, VStone Mecanumrover3

Control and Optimization: model predictive control, trajectory optimization, kinematics, Kalman filtering, convex optimization, CVXPY, ADMM

Programming and Systems: Python, C++, CUDA, Linux, Git, C, Bash, GPU optimization, multithreading, quantization, ExecuTorch, Vulkan, MATLAB, JavaScript, Java

Soft Skills: Problem-solving, adaptability, cross-functional collaboration, technical communication, teamwork, project ownership, mentoring, and time management.

Extracurricular: Guitar, drums, and saxophone playing, musical improvisation, strength training, cooking

Languages: English (fluent), German (conversational), Spanish (native)